

Week 5 — Percentage, Ratio & Proportion, Profit/Loss & Simple Interest — Answer Key

Q1. C) 70 — $20\% \text{ of } 350 = 0.20 \times 350 = 70$.

Q2. B) 40% — $2/5 = 0.4 = 40\%$.

Q3. B) 25% — $45/180 \times 100 = 25\%$.

Q4. C) 36 — 5 parts = 45 boys, so 1 part = 9; girls = 4 parts = 36.

Q5. C) ₹360 — Total parts = 5, so 1 part = ₹120; B's share = 3 parts = ₹360.

Q6. C) ₹105 — Cost per pen = ₹45 ÷ 3 = ₹15; 7 pens = 7 × ₹15 = ₹105.

Q7. B) 20% — Profit = ₹300 – ₹250 = ₹50; profit % = $50/250 \times 100 = 20\%$.

Q8. B) 20% — Loss % = $40/200 \times 100 = 20\%$.

Q9. C) ₹1,200 — SI = $(\text{Principal} \times \text{Rate} \times \text{Time}) \div 100 = (5000 \times 8 \times 3) \div 100 = ₹1,200$.

Q10. B) 5% — Rate = $(\text{SI} \times 100) \div (\text{Principal} \times \text{Time}) = (500 \times 100) \div (2000 \times 5) = 5\%$.

Q11. C) 288 — $60\% \text{ of } 480 = 0.6 \times 480 = 288$.

Q12. C) 10% — Loss = ₹450 – ₹405 = ₹45; loss % = $45/450 \times 100 = 10\%$.

Q13. C) 77 — Total parts = 4+7 = 11, so 1 part = $121 \div 11 = 11$; larger number = 7 × 11 = 77.

Q14. C) ₹9,600 — SI = $(8000 \times 10 \times 2) \div 100 = ₹1,600$; amount = 8000 + 1600 = ₹9,600.

Q15. B) ₹400 — Marked price = $400 \times 1.25 = ₹500$; after a 20% discount: $500 \times 0.8 = ₹400$. The selling price ends up exactly equal to the cost price — a 25% markup followed by a 20% discount cancel out exactly ($1.25 \times 0.8 = 1$), so there's neither profit nor loss. This is a classic "trick" to watch for: a bigger-sounding percentage off doesn't always mean a lower final price than you started with.

Scoring: 15/15 — excellent, move to Week 6. 13–14/15 — solid, review the missed ones. Below 13/15 — these are brand-new topics, so a lower score here is completely normal the first time. Re-read each worked explanation above, then redo the assignment in 2 days rather than pushing ahead.
